CS 7545 F26, Homework 3

100 point assignment.
110 points possible.
Points above 100 are divided by five and become bonus points.
Maximum bonus 2 points.

Due November 12, 2026

Exercise 1 (35 points)

In this exercise we study a one-dimensional estimation problem in which the population loss can have a degenerate Hessian at its minimizer, or more than one global minimizer, and we examine how each of these affects empirical risk minimization.

Fix $\lambda \geq 0$. Data are drawn as $z \sim \mathcal{N}(0, 1)$ and for a parameter $\theta \in \mathbb{R}$ we use the regularized squared loss

$$\ell(z, \theta) = (z - \theta)^2 + \lambda(\theta^2 - 1)^2. \tag{1}$$

The population loss is $L(\theta) = \mathbb{E}_z[\ell(z, \theta)]$. Given i.i.d. samples $S = \{z_1, \ldots, z_n\}$, the empirical loss is $\hat{L}_S(\theta) = \frac{1}{n} \sum_{i=1}^{n} \ell(z_i, \theta)$, and $\hat{\theta}_S \in \arg\min_{\theta \in \mathbb{R}} \hat{L}_S(\theta)$ denotes an empirical risk minimizer (ERM).

Part A (5 points)

Show the following,

$$L(\theta) = 1 + \theta^2 + \lambda(\theta^2 - 1)^2 \tag{2}$$

$$\hat{L}_S(\theta) = \frac{1}{n} \sum_{i=1}^{n} z_i^2 - \bar{z}_S^2 + (\theta - \bar{z}_S)^2 + \lambda(\theta^2 - 1)^2 \tag{3}$$

with $\bar{z}_S = \frac{1}{n} \sum_{i=1}^{n} z_i$.

Part B (15 points)

With $\lambda = 1/2$,

1. Show that $\theta^\star = 0$ and $L''(\theta^\star) = 0$.
2. Show that the ERM is unique and equal to $\hat{\theta}_S = (\bar{z}_S)^{1/3}$ (the real cube root).
3. Use Jensen's inequality to show that $\mathbb{E}[L(\hat{\theta}_S) - L(\theta^\star)] \geq \frac{1}{2}\left(\mathbb{E}[|\bar{z}_S|]\right)^{4/3}$.
4. Use the fact that $\mathbb{E}_{u\sim\mathcal{N}(0,\sigma^2)}[|u|] = \sqrt{\frac{2}{\pi}}\sigma$ to show that the expected excess loss is $\Omega(n^{-2/3})$. Note this is worse than the $\mathcal{O}(n^{-1})$ rate for non-degeneracy.

Part C (15 points)

With $\lambda = \frac{2}{3}$,

1. Show that L has two global minimizers $\theta^\star = \pm\frac{1}{2}$.
2. Ignoring the probability zero event $\bar{z}_S = 0$, show that any ERM $\hat{\theta}_S$ has the same sign as $\bar{z}_S$.

   Hint: first show that $\hat{L}_S(\theta) - \hat{L}_S(-\theta) = -4\bar{z}_S\theta$.
3. Show that any ERM $\hat{\theta}_S$ satisfies

$$4\hat{\theta}_S^3 - \hat{\theta}_S = 3\bar{z}_S \tag{4}$$

   and use this together with the previous part to show that

$$\left|\hat{\theta}_S - \tfrac{1}{2}\operatorname{sign}(\bar{z}_S)\right| \leq \tfrac{3}{2}\left|\bar{z}_S\right| \tag{5}$$

   Hint: $\theta(4\theta^2 - 1) < 0$ for $0 < \theta < \frac{1}{2}$.
4. Using the Gaussian tail bound $\Pr(|u| \geq t) \leq 2\exp\left(-\frac{t^2}{2\sigma^2}\right)$ (for $u \sim \mathcal{N}(0,\sigma^2)$), show that for any $\delta \in (0,1)$, with probability $1-\delta$,

$$\left|\hat{\theta}_S - \tfrac{1}{2}\operatorname{sign}(\bar{z}_S)\right| \leq \tfrac{3}{2}\sqrt{\frac{2\ln(2/\delta)}{n}} \tag{6}$$

   Since $\Pr(\bar{z}_S > 0) = \Pr(\bar{z}_S < 0) = \frac{1}{2}$, note that $\hat{\theta}_S$ does not converge in probability to any point as $n \to \infty$ (no need to show).

Exercise 2 (15 points)

In this exercise, we show a way to construct a pseudo-metric from a kernel. Here we define a "pseudo-metric" which satisfies all the conditions of a metric (distance function) except for $d(x, y) = 0 \implies x = y$.

Definition 1 (Pseudo-metric)**.** *A function $d : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is a pseudo-metric if:*

- $d(x, x) = 0$
- $d(x, y) = d(y, x)$
- $d(x, y) \leq d(x, z) + d(z, y)$

For a PSD kernel function k, prove that $d(x, y) = \sqrt{k(x, x) - 2k(x, y) + k(y, y)}$ is a pseudo-metric.

Hint for the triangle inequality: for a set of three points $\{x_i\}$, define $K \in \mathbb{R}^{3\times 3}$ with $K_{i,j} = k(x_i, x_j)$, then find vectors v such that $d(x_i, x_j) = \sqrt{v^{\mathrm{T}} K v}$.

Exercise 3 (20 points)

In this question, we calculate the prediction for kernel ridge regression in a particular setting.

Suppose our input space $\mathcal{X}$ can be partitioned ("clustered") into an infinite union of sets $\mathcal{X}_j \subset \mathcal{X}$ such that $\mathcal{X}_j \cap \mathcal{X}_{j'} = \emptyset$ if $j \neq j'$, and $\mathcal{X} = \bigcup_{j=1}^{\infty} \mathcal{X}_j$. Define the kernel $k(x, x') = \mathbf{1}[\exists j : x \in \mathcal{X}_j \wedge x' \in \mathcal{X}_j]$. In other words, the kernel function is 1 for points in the same "cluster" and 0 otherwise.

For a training set $\{(x_i, y_i)\}_{i=1}^n$, show that the kernel ridge regression (minimizing $\frac{1}{n}\sum_{i=1}^n (h(x_i) - y_i)^2 + \lambda \|h\|^2$) for a prediction on a test point x_0 is

$$\frac{\sum_{i \in C(x_0)} y_i}{|C(x_0)| + n\lambda} \tag{7}$$

where $C(x_0) = \{i : k(x_0, x_i) = 1\}$.

Hint: you may find it helpful to note that the inverse of a block-diagonal matrix is also a block-diagonal matrix with the matrix inverse of the blocks, and to use the Sherman–Morrison formula to evaluate the inverse of the sum of a diagonal matrix and a rank-one matrix.

Exercise 4 (20 points)

In this exercise, we show that we can use the entropy of the prior over a hypothesis class rather than the log size in a realizable generalization bound.

Suppose we have set of inputs $\mathcal{X}$ and a finite hypothesis class $H \subset \{h : \mathcal{X} \to \{-1, 1\}\}$. Suppose we are in a Bayesian setting with a prior π over H and a distribution $\mathcal{D}$ over $\mathcal{X}$. Thus,

$$h^\star \sim \pi \tag{8}$$
$$x_i \sim \mathcal{D} \tag{9}$$
$$y_i = h^\star(x_i) \tag{10}$$

For a dataset $S \subset \mathcal{X} \times \{-1, 1\}$ of size $|S| = n$, define $\hat{L}_S(h) = \frac{1}{n} \sum_{(x,y)\in S} \mathbf{1}[h(x) \neq y]$. Also, define $L_{h^\star}(h) = \Pr_{x\sim\mathcal{D}}(h(x) \neq h^\star(x))$ as the population loss conditioned on $h^\star$. Define

$$\hat{h}_{\text{MAP}}(S) \in \arg\max_{h\in H} \Pr(h = h^\star|S) \tag{11}$$

1. Show that conditioned on $h^\star$, for any hypothesis h, $\Pr(\hat{L}_S(h) = 0) \leq \exp(-nL_{h^\star}(h))$.
2. Show that for any $\delta \in (0, 1)$, conditioned on $h^\star$, for any hypothesis h, with probability $1 - \pi(h)\delta$, either $\hat{L}_S(h) > 0$ or $L_{h^\star}(h) \leq \frac{\ln(1/\pi(h))+\ln(1/\delta)}{n}$.
3. Show that conditioned on $h^\star$, with probability $1 - \delta$ (over sampling S),

$$L_{h^\star}(\hat{h}_{\text{MAP}}(S)) \leq \frac{\ln(1/\pi(\hat{h}_{\text{MAP}}(S))) + \ln(1/\delta)}{n} \tag{12}$$

4. Show that,

$$\Pr(L_{h^\star}(\hat{h}_{\text{MAP}}(S)) \geq \epsilon|h^\star) \leq \frac{\exp(-n\epsilon)}{\pi(h^\star)} \tag{13}$$

5. Show that,

$$\mathbb{E}_S[L_{h^\star}(\hat{h}_{\text{MAP}}(S))|h^\star] \leq \frac{1 + \ln(1/\pi(h^\star))}{n} \tag{14}$$

6. Finally, show that for independently sampling a training set $S = \{(x_i, y_i)\}_{i=1}^n$ and a test point (x_0, y_0),

$$\Pr(\hat{h}_{\text{MAP}}(S)(x_0) \neq y_0) \leq \frac{1 + \text{Entropy}(\pi)}{n} \tag{15}$$

where the entropy of the distribution π is $\sum_{h\in H} -\pi(h) \ln(\pi(h))$.

Exercise 5 (20 points)

In this exercise, we compute the Laplace approximation for Bayesian Linear and Logistic Regression.

1. For Bayesian Linear Regression and fixed data $\{x_i\}_{i=1}^n$,

$$\theta^\star \sim \mathcal{N}(0, \lambda I_d) \tag{16}$$
$$y_i \sim \mathcal{N}(\theta^\star \cdot x_i, \sigma^2) \tag{17}$$

   Write the posterior density $\Pr(\theta^\star = \theta | \{(x_i, y_i)\}_{i=1}^n)$ in the form of $\exp(-E(\theta))/Z$ for a constant Z (that you don't need to compute).

2. Derive the Laplace approximation for $\theta^\star | \{(x_i, y_i)\}_{i=1}^n$. You should compute both the mean and covariance in terms of λ, σ^2, and $\{(x_i, y_i)\}_{i=1}^n$.

   Fun fact: For Bayesian Linear Regression, the Laplace approximation is exact (i.e. it's not an approximation)!

3. For Bayesian Logistic Regression and fixed data $\{x_i\}_{i=1}^n$,

$$\theta^\star \sim \mathcal{N}(0, \lambda I_d) \tag{18}$$
$$y_i \sim \begin{cases} 1 & \text{w.p. } \sigma(\theta^\star \cdot x_i) \\ -1 & \text{w.p. } \sigma(-\theta^\star \cdot x_i) \end{cases} \tag{19}$$

   where $\sigma(u) = \frac{1}{1+\exp(-u)}$ is the Sigmoid function.

   Write the posterior density $\Pr(\theta^\star = \theta | \{(x_i, y_i)\}_{i=1}^n)$ in the form of $\exp(-E(\theta))/Z$ for a constant Z (that you don't need to compute).

4. Derive the Laplace approximation for $\theta^\star | \{(x_i, y_i)\}_{i=1}^n$. You should compute both the mean and covariance in terms of $\hat{\theta}_{\text{MAP}}$, λ, and $\{(x_i, y_i)\}_{i=1}^n$, where $\hat{\theta}_{\text{MAP}} = \arg\max_\theta \Pr(\theta^\star = \theta | \{(x_i, y_i)\}_{i=1}^n)$.

   Hint: you can use the following facts for $\ell(\theta, x, y) = -\ln \sigma(yx \cdot \theta)$,

$$\nabla_\theta \ell(\theta, x, y) = -y\sigma(-yx \cdot \theta)x \tag{20}$$
$$\nabla_\theta^2 \ell(\theta, x, y) = \sigma(x \cdot \theta)\sigma(-x \cdot \theta)xx^{\mathrm{T}} \tag{21}$$